

Project Proposal & System Blueprint

App Name: MyNest

Motto: *Before your memory fades away. ---*

1. The Problem We Are Solving

Human memory is the most fragile artifact we possess. Every single day, irreplaceable pieces of our personal history silently fade into oblivion. The exact sound of a grandfather's laugh, the unspoken struggles of a mother's youth, the context behind a faded black-and-white photograph—these are the true treasures of human life.

When a loved one passes away, or when time simply takes its toll on their mind, an entire library of history, emotion, and wisdom burns down with them. We are left with silent, disconnected photos and fragmented recollections that lose their meaning over generations. The true tragedy is not just the loss of life; it is the silence that follows. There is a profound, emotional need to capture the soul of these moments, the raw voices, and the genuine stories from our elders and loved ones before time runs out. We need a sanctuary for our history.

The Real-Life Scenario: Imagine finding a black-and-white photo of your late grandfather from 1970. He is no longer here to tell you about it. You need a place to securely save that photo and record *your own* memories of what he told you about that day.

Right now, families usually just have scattered photos in WhatsApp groups. Existing apps are often cluttered photo galleries that treat memories like ordinary files. We need a system that feels personal, minimal, and focuses purely on preserving the *story*. Furthermore, for relatives who are still with us, it is difficult to ask an 80-year-old to download a new application. We need a hybrid system: a powerful, structured vault for the main user, and accessible web links for older relatives to easily contribute.

2. Project Goal & Design Philosophy

MyNest is a dedicated digital sanctuary designed to preserve family history across generations, supported by AI processing.

The Design Philosophy: The ultimate selling point of this app is its database visualization and emotional resonance. The UI must be minimal, using clean whitespace, warm tones, and soft rounded corners. It should not feel like a tech app or a complex social network; it should feel like a private digital museum—a place of reverence and connection.

The core goal is to take scattered family memories (voice notes, old photos) and structure

them into a readable **Digital Story Book**, an interactive **Family Tree**, and an auto-generated **Historical Timeline**.

3. Technology Stack

- **Mobile App:** Flutter & Dart (For a smooth, cross-platform mobile UI).
- **Web Pages:** Simple HTML/CSS/JavaScript (So relatives can click a link and open a webpage instantly).
- **Cloud Backend & Database:** A ready-made cloud database system to handle authentication and host data securely.
- **AI Engine:** Google Gemini API (Used to transcribe audio, structure story chapters, extract historical dates, and map family tree connections from text input).

4. The Core Concept: In-App Studio & Smart Links

To build the family vault logically, MyNest uses two main methods:

Method 1: The In-App Memory Studio (For the Main User & Deceased Relatives)

The app has a built-in studio. If you find a photo of a relative who has passed away, you simply go to their profile in the app, click "Add Memory," upload the photo, and record your own voice note remembering them.

Method 2: The Smart Web Links (For Living Relatives)

1. **The "Add Memory" Link:** You send this to your aunt. She clicks it, talks into her phone browser, and submits it.
2. **The "Photo Context" Link:** You upload an old photo and send the link to the family group chat to ask, "Who is in this picture?"

Smart Auto-Population & Timelines: When a relative uses a web link, they type their relation (e.g., "I am Mursalin's mother's sister"). **The AI analyzes the relation text, converts it into structured relationship data, and sends it to the database as a pending family node for user approval. It simultaneously extracts dates and events (e.g., "1970 - Grandfather's Wedding") to automatically construct a master historical timeline.**

5. UI Screen Design & Logical App Flow

This section details the architectural purpose and required components for each screen. UI designers must use this robust framework to construct the app's visual interfaces.

Screen 1: Login & Sign Up (App)

- **Description:** This screen serves as the secure entry point for the MyNest application, ensuring that private family data remains protected against unauthorized access. It provides the foundational authentication mechanisms required to either create a new

family vault or access an existing one safely. Users can register for a new account using their email credentials to initiate a fresh database instance in the cloud. Alternatively, returning users can securely log in to synchronize and retrieve their previously stored family history and hierarchical tree data. The interface is designed to be minimal and welcoming, establishing a tone of security immediately upon launching the app. By handling secure authentication first, the system guarantees that all subsequent CRUD operations and data retrievals are safely scoped exclusively to the authorized user profile.

- **Required UI Components:** * Large, clean "MyNest" Logo at the top.
 - Two standard Text Fields: Email and Password.
 - Two prominent Buttons: "Log In" and "Create Account".

Screen 2: Initial Profile Setup (App - One Time Only)

- **Description:** The Initial Profile Setup screen is a mandatory onboarding step required to initialize the family database architecture correctly. It functions as the primary data collection point for the "root node" of the family tree, anchoring the entire relational structure to the main account holder. Users are prompted to input their foundational biographic data, including their full legal name and exact date of birth via a calendar interface. They can also upload a primary profile photograph which will visually represent them globally across the application's interactive tree and memory timelines. By completing this screen, the user successfully instantiates the core database collections needed for the system's relational mapping to function. This step critically ensures the backend has the necessary spatial anchor point before any external relatives are invited to contribute data.
- **Required UI Components:**
 - ☐ Title: "Let's start your Family Tree."
 - ☐ Large circular Photo uploader block.
 - ☐ Text Fields: "Your Full Name", "Date of Birth" (Date Picker).
 - ☐ Bottom Button: "Start My Family Vault".

Screen 3: The Dashboard (App - Bottom Nav Tab 1)

- **Description:** The Dashboard operates as the central command hub of the application, aggregating system status, quick actions, and recent data inputs. It provides a comprehensive overview of the family vault's growth through quick statistics, tracking total members, compiled memories, and generated story chapters. Users utilize this screen to monitor incoming memory submissions via a dynamic activity feed and to track the overall completion status of the AI Story Book. The primary functional utility of this screen is the generation of smart web links, allowing users to instantly dispatch contribution requests to external relatives. Furthermore, it serves as the direct access portal to the compiled Story Book and offers a seamless pathway to the In-App Memory Studio for manual data entry. This centralized architecture ensures that users can manage, monitor, and expand their entire digital heritage from a single, highly informative interface.
- **Required UI Components:**

Top App Bar: "Welcome back, [First Name]".

Hidden Teacher-Demo Toggle: A hidden "Run Demo" toggle in the header.

Mission Banner (Empty state): "Share one memory today — tap + to open the In-App Studio or send a link."

Quick Stats & Goals Row: "Members: 12 • Memories: 28 • Chapters: 5" with a subtle progress bar: "Tree completion: 40%".

The AI Story Book Section: Graphic of a book cover. Status: "📖 8 Chapters | 🔔 2 new memories". Button: [Open Book].

Gather Memories (Link Generation):

- Buttons: [+ Add Memory Link], [Photo Context Link], [Vault Share Link].
- *Link Generation Flow:* When creating a Photo Context Link, the user must first upload or select an existing photo. The system then generates a shareable link associated with that image.
- *Feedback Modal:* Header: "Link Created". Subtext: "Share this link with anyone (no app required)." Buttons: [Copy], [Share], [Set Expiration].

Recent Activity (Filtered Feed): Filters: [All], [New], [Needs Review]. Feed Items show icon, title, name, time. Badges: [RAW], [AI-READY], [CHAPTERED].

Screen 4: The In-App Memory Studio (App - Sub-screen)

- **Description:** The In-App Memory Studio is a dedicated data entry environment optimized for the primary account holder to manually digitize family history. It is specifically designed to handle memories related to deceased relatives or physical artifacts that cannot be processed via external web links. Users can utilize this studio interface to upload scanned photographs, explicitly tag specific family members from their database, and attach rich contextual data. The screen provides dual input modalities, allowing users to either type detailed textual narratives or record high-fidelity voice notes directly into the system. Once submitted, the backend securely stores the media assets and automatically routes the narrative data to the Gemini AI engine for transcription and contextual formatting. This screen ultimately guarantees that the core user has absolute control over populating the family vault without relying solely on external contributors.
- **Required UI Components:**
 - ☐ Image Uploader box.
 - ☐ Dropdown selector: "Tag a Family Member".
 - ☐ Two massive interaction buttons: [🎤 Record Audio] or [🖋️ Type Story].
 - ☐ [Save to Vault] submit button.

Screen 5: The Story Book Reader (App - Sub-screen from Dashboard)

- **Description:** The Story Book Reader is a premium, distraction-free consumption interface designed to display the AI-compiled family history. It transforms fragmented, raw

database entries into a cohesive, chapter-based narrative structure heavily resembling a published digital e-book. Users navigate to this screen to seamlessly read the polished stories generated by the Gemini engine based on the cumulative multimedia inputs of their relatives. The interface prioritizes typographic readability, pairing high-resolution archival photography with beautifully formatted, justified text blocks. Additionally, the screen features an integrated text-to-speech accessibility function, allowing users to listen to the entire family history narrated aloud programmatically. This screen represents the ultimate value proposition of the application, converting raw data collection into a meaningful, easily digestible legacy artifact.

- **Required UI Components:**

- ☐ Top header block: Expanded old family photo.
- ☐ Middle body: Chapter title and AI-written story paragraphs.
- ☐ Navigation controls: [< Previous] and [Next >] buttons at the bottom.
- ☐ Floating Action Button: [ Play Audio] (triggers text-to-speech functionality).

Screen 6: The Vault / Raw Timeline (App - Bottom Nav Tab 2)

- **Description:** The Vault functions as the chronological master ledger of all raw data points inputted into the family application based on the date they were uploaded. It displays an unfiltered, vertically scrolling timeline of every photograph, text entry, and voice note submitted by family members across all platforms. Users leverage this screen to verify recent uploads, search for specific historical events using keyword queries, and review the AI processing status of individual memories. By tapping on any item in the feed, users can access a detailed management view to execute critical update or delete operations on the specific data node. This provides total administrative control, ensuring that erroneous submissions, typos, or duplicate files can be swiftly and permanently managed. The Vault ultimately serves as the transparent, unedited archive complementing the polished, AI-generated Story Book experience.

- **Required UI Components:**

- ☐ Top App Bar: "Raw Memories Vault" with a functional search icon.
- ☐ Main Screen: A vertical feed of "Memory Cards" featuring status indicator badges.
- ☐ **Interaction (Memory Detail View):** Tapping a memory card opens the full view. This view includes action buttons to [Edit Story], [Delete Memory], and [Tag Person].
- ☐ Floating + Button to quickly launch the In-App Studio.
- ☐ Secondary Action Button: [View Historical Timeline] to route to Screen 6A.

Screen 6A: The Historical Timeline (App - Sub-screen from Vault)

- **Description:** The Historical Timeline serves as a dynamic, auto-generated chronological ledger of the family's most significant historical milestones. Unlike the standard Vault which sorts by upload date, this interface leverages the AI Engine to extract spoken or written dates from memories and map them onto a linear historical axis. Users can scroll through decades (e.g., 1970, 1985, 2002) to see major family events like weddings, house constructions, or graduations laid out sequentially. It provides a macro-level, bird's-eye

view of the family's legacy spanning multiple generations. This visualization transforms scattered data points into a cohesive historical record, making it instantly comprehensible. Users can tap any milestone on the timeline to directly access the associated full memory, photographs, and audio recordings. This feature elevates the application from a simple storage vault into a true historical archiving and visualization tool.

- **Required UI Components:**

- ☐ Top App Bar: "Historical Timeline".
- ☐ Vertical Axis Line with Decade Markers (e.g., "1970s", "1980s").
- ☐ Milestone Cards attached to the axis (Year + Event Title + Thumbnail Image).
- ☐ Filter Dropdown: "Filter by Person" or "Filter by Event Type".

Screen 7: Family Tree Builder (App - Bottom Nav Tab 3)

- **Description:** The Family Tree Builder is a highly interactive data visualization canvas that maps complex relational database entries into an intuitive graphical interface. It provides users with a macro-level perspective of their ancestry, utilizing a node-based architectural design to represent individual family members and their connections. Users navigate this expansive digital canvas using standard touch gestures like pinching to zoom and panning to explore different generational branches efficiently. Crucially, this screen serves as the primary resolution center for pending connections, allowing users to review, approve, or reject nodes automatically generated by the AI from web links. Furthermore, it acts as the launching point for manual database expansion, enabling users to add parents, children, or partners directly to the visual map via tap interactions. This screen ensures that intricate familial relationships are both easily understood visually and effortlessly managed functionally.

- **Required UI Components & Interactions:**

- ☐ **Navigation Controls:** Pinch to zoom, drag to pan. Top-right controls: [-], [+], [Center on Me].
- ☐ **Persistent Legend:** ● Living, ○ Deceased, ◐ Pending, — Friend.
- ☐ **Pending Connections UX:** Banner: "🔔 New connection detected. [Review Pending]". Review Flow Modal: "This will add Shirin as your mother's sister." -> Actions: [Approve], [Edit & Approve], or [Reject].
- ☐ **Inline Node Preview:** Tapping a node opens a quick-profile preview bubble:
 - Content: "Aunty Shirin • mother's sister".
 - Buttons: [View Profile], [Add Parent], [Add Child], [Add Partner], [Add Sibling].

Screen 7A: Add / Edit Person Form (App - Sub-screen from Tree)

- **Description:** This specialized sub-screen provides the structured input form necessary for creating and modifying individual profiles within the relational database. It ensures that all biographical data entering the system is cleanly formatted, validated, and strictly categorized for seamless tree generation. Users access this interface to manually input a new relative's full name, exact date of birth, gender, and their specific relational vector to existing nodes. It also functions as the primary update mechanism, allowing users to

correct typographical errors or seamlessly update profile photographs for previously established family members. By enforcing a standardized data entry protocol, this screen prevents database corruption and ensures the visual family tree remains accurate and unbroken. The streamlined UI design minimizes friction while successfully capturing all required metadata for the backend system to map the family correctly.

- **Required UI Components:**

- ☐ Circular photo uploader at the top.
- ☐ Text Field: "Full Name".
- ☐ Date Picker: "Date of Birth".
- ☐ Dropdown: "Gender".
- ☐ Relationship selector dropdown: Parent, Child, Partner, Sibling, Friend.
- ☐ Optional Text Field: "Notes".
- ☐ Action Buttons: [Save] and [Cancel].

Screen 8: Family Member Profile (App - Sub-screen from Tree)

- **Description:** The Family Member Profile operates as a dedicated digital memorial and data aggregation hub for a specific individual within the database. It isolates and retrieves only the photographs, voice notes, and stories explicitly associated with a unique person's ID, filtering out the rest of the global timeline. Users utilize this screen to deeply explore the life of a single relative, viewing their media through an organized gallery or reading their specific narrative contributions. It also provides advanced administrative capabilities, including a toggle to mark a profile as deceased, which locks destructive actions to prevent accidental data loss. Furthermore, users can execute complex merge operations here to resolve duplicate profiles or permanently delete an individual from the system hierarchy entirely. This screen effectively transforms a single database node into a rich, interactive, and highly manageable multimedia biography.
- **Required UI Components:**
 - ☐ **Top Header:** Cover photo, profile picture, Name, Date of Birth.
 - ☐ **Profile Toggles:** [Mark as Deceased] toggle (disables accidental deletion and turns node grayscale).
 - ☐ **Merge Tool:** [Merge Duplicate] option.
 - ☐ **Action Row:** [+ Add Memory for Them], [Edit Info], [Delete Person].
 - ☐ **Tabbed Menu:** [Stories] and [Photo Gallery].

Screen 9: My Profile & Settings (App - Bottom Nav Tab 4)

- **Description:** The Profile and Settings screen provides the essential account management interface for the primary application owner. It acts as the control center for global user preferences, authentication modifications, and overarching database import operations. Users navigate here to execute routine updates to their personal credentials, such as modifying their display name or securely changing their access passwords via the backend provider. Importantly, this screen houses the data import functionality, allowing

users to paste a Vault Share Link to entirely overwrite or append their local database with an external family tree. It also contains the secure logout mechanism to terminate the active session token and protect local data access on the device. This interface ensures the user maintains complete sovereignty over their software environment, security protocols, and data portability.

- **Required UI Components:** * Header displaying active Email, Name, and Profile Photo.
 - ☐ Settings List Buttons: [Edit Details], [Paste a Vault Share Link] (to import a tree), [Log Out].

Screen 10: The Web Portals (Web Platform for Relatives)

- **Description:** The Web Portals represent the frictionless, external data collection gateways designed specifically for non-technical or elderly relatives. By operating entirely within a standard mobile web browser, these portals completely eliminate the friction of app store downloads, account registrations, and complex onboarding flows. Users interacting with these screens are guided through an incredibly simplified, linear submission process utilizing oversized UI elements and high-contrast typography. They can effortlessly upload archival imagery from their device gallery and utilize a massive, universally understood microphone button to record high-fidelity voice narratives. The system then automatically structures this external input, bundling the audio, imagery, and stated relationship data before securely transmitting it to the cloud database. This architectural decision is the cornerstone of the application, ensuring maximum data collection from older generations before those memories are permanently lost.
- **Required UI Components:** * Accessible text fields: "Your Name", "How exactly are you related to the user?".
 - ☐ Media Button: [Upload Photo (Optional)].
 - ☐ A giant, unmissable microphone button: **"Tap to Record your Story"**.
 - ☐ A clear, high-contrast "Submit" button.

6. The 7 Required CRUD Pages (Grading Rubric Mapping)

The app design strictly follows the requirement of having at least 7 unique functional pages that perform CRUD (Create, Read, Update, Delete) operations.

1. Authentication / Setup Page (Screens 1 & 2)

CRUD: *Create* (Inserts a new user record and creates the root family node).

2. Dashboard & Link Generator (Screen 3)

CRUD: *Create* (User generates unique share links) & *Read* (Fetches filtered recent activity).

3. In-App Studio (Screen 4)

CRUD: *Create* (User directly uploads their own photos/audio to the database).

4. The Vault Timeline & Historical Timeline (Screens 6 & 6A)

CRUD: *Read* (Retrieves entries from the database and maps them chronologically), *Update* (Edit stories), & *Delete* (Remove memory).

5. Family Tree Canvas (Screen 7)

CRUD: *Read* (Queries relationship database to draw nodes) & *Update* (Approves pending connections).

6. Add/Edit Person Form (Screen 7A)

CRUD: *Create* (Adds new node) & *Update* (Modifies existing person's details).

7. Family Member Profile (Screen 8)

CRUD: *Delete* (Removes a person entirely) & *Read* (Filters the Vault by PersonID).

8. The Web Contribution Portals (Screen 10)

CRUD: *Create* (Relatives insert new audio files/text) & *Update* (Append comments to photo contexts).

7. Small Demo Checklist

Use this predictable sequence to flawlessly demonstrate the app's capabilities during the final lab presentation:

1. **Onboarding:** Create an account → Show the Dashboard empty state with the mission banner.
2. **Link Generation:** Tap [Photo Context Link] → Upload a photo → Show the feedback modal.
3. **Web Submission:** Submit a fake memory/photo via the copied web link → See the new item appear in the Recent Activity feed.
4. **Tree & Timeline Resolution:** Open the Family Tree → Highlight the Pending Connection alert; review the AI-generated relationship sentence and click [Approve]; watch the node appear. Then jump to the Historical Timeline to show the AI-extracted dates.
5. **The Vault Editing:** Open the Vault → Open the Memory Detail View → [Edit] the story text to demonstrate update functionality.
6. **Profile Interaction:** Tap the new node in the tree → Open the preview bubble → Click [View Profile] → Add a memory directly to a deceased person via the In-App Studio.
7. **CRUD Deletion:** Delete a memory from the profile and click the [Undo] toast to demonstrate safety features.